What the Second Congruence Buys: Quantified Evidence on Popovych's Conjecture

(standalone note — v0.1)

Dan Shumow Claude (Anthropic Fable 5)

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Abstract

Agrawal's conjecture asserts that for a prime $r \nmid n$ the congruence $(X-1)^n \equiv X^n - 1 \pmod{n, X^r-1}$ forces n prime or $n^2 \equiv 1 \pmod r$; Popovych (2009) conjectured that adding the second congruence $(X+2)^n \equiv X^n + 2$ repairs the statement, and beyond exhaustive search ($n < 10^{17}$, Primaboinca) no evidence has been published in either direction. We quantify what the second section costs the only known counterexample engine, the Lenstra–Pomerance Frobenius-matching mechanism. The mechanism's cost for the single section $X-1$ is governed by an exact *inversion collapse* available because the zero $\{1\}$ of $X-1$ is Galois-stable; we prove the zero -2 of $X+2$ has trivial stabilizer — for every $\gamma \neq 1$ the ratio $(\zeta_r^\gamma + 2)/(\zeta_r + 2)$ is not even a unit of $\mathbb{Z}[\zeta_r]$ — so no collapse exists for the second section at any depth. Consequently the pair mechanism must make a product of cyclotomic torus orders of total dimension $\min\{d : d \mid r-1,\ d \geq 3\} \geq 3$ simultaneously smooth at every pool prime, for *every* modulus r, against the single section's optimum of 2; at Popovych's own $r = 5$ the burden is 4 ($p-1$, $p+1$ and p^2+1 simultaneously smooth), a Dickman tax of 10^{-7}–10^{-11} at representative parameters that empties the measured supply of the corresponding Agrawal-directed harvest. The burden lower bounds are unconditional given the collapse identities; that they are *attained* (so that no thinner collapsed pool rescues the mechanism) is an Artin-type genericity hypothesis, stated explicitly and supported by exact-order computations with zero exceptions. To our knowledge this is the first quantified evidence distinguishing Popovych's conjecture from Agrawal's beyond search ranges.

1 Introduction

For coprime n, r write $T(a, n, r)$ for the congruence $(X+a)^n \equiv X^n + a \pmod{n,\ X^r-1}$. Agrawal's conjecture [1, 2] states that $T(-1, n, r)$ with r prime, $r \nmid n$, forces n prime or $n^2 \equiv 1 \pmod r$; it is open, and believed false on the strength of the Lenstra–Pomerance (L–P) heuristic construction [3]. Popovych [4] proved a generalization of the L–P counterexample mechanism and proposed the repaired conjecture:

Conjecture 1 (Popovych)**.** *If r is a prime not dividing n and both $T(-1, n, r)$ and $T(2, n, r)$ hold, then n is prime or $n^2 \equiv 1 \pmod r$.*

Its evidence base is thin in both directions: a plausibility argument that the pair $\{X-1, X+2\}$ generates a large unit subgroup [4, 5], and the Primaboinca exhaustive search ($10^{10} < n < 10^{17}$, nothing found). This note supplies the first quantitative statement: *every known counterexample mechanism pays strictly more against the pair than against the single section — at least one extra algebraic-torus dimension of smoothness burden globally, and two extra at the canonical modulus $r = 5$.*

The engine behind all published counterexample approaches [3, 4, 7] is *Frobenius matching*: modulo each prime $p \mid n$, the congruence $T(a, n, r)$ holds if $n \equiv p^j \pmod{\operatorname{lcm}(r, \operatorname{ord}(X+a))}$, where the order is taken in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$, because $(X+a)^{p^j} \equiv X^{p^j} + a$ exactly (Frobenius and Fermat). A construction assembles a squarefree $n = p_1 \cdots p_k$ meeting these congruences at every p_i simultaneously, which requires the controlling orders to be smooth enough to fit a common modulus. The cost of the construction is therefore governed by the *support* of those orders, and the support is where the two sections differ.

2 The collapse for $X-1$ and its absence for $X+2$

Fix an odd prime $r \neq 3$, a prime $p \nmid 3r(2^r+1)$, and let $d = \operatorname{ord}_r(p)$. Write $\rho_r(p)$ and $\rho'_r(p)$ for the multiplicative orders of $X-1$ and $X+2$ in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$; both divide $p^d - 1 = \prod_{e \mid d} \Phi_e(p)$, a factorization indexed by *cyclotomic levels* e, with $\Phi_e(p)$ the order of the e-th cyclotomic torus over $\mathbb{F}_p$, of dimension $\varphi(e)$.

Lemma 2 (inversion collapse for the first section)**.** *If d is even, then $(X-1)^{p^{d/2}-1} = -X^{-1}$ in $\mathbb{F}_p[X]/\Phi_r(X)$, and hence $\rho_r(p) \mid 2r(p^{d/2}-1) = 2r \prod_{e \mid d/2} \Phi_e(p)$.*

Proof. $(\mathbb{Z}/r)^\times$ is cyclic, so its unique involution is -1 and $p^{d/2} \equiv -1 \pmod r$. In each field component, with $X \mapsto \zeta$ a primitive r-th root of unity, iterated Frobenius gives $(\zeta-1)^{p^{d/2}} = \zeta^{-1} - 1 = -\zeta^{-1}(\zeta-1)$; divide by $\zeta - 1 \neq 0$. Finally $(-X^{-1})^{2r} = 1$ since $X^r \equiv 1 \bmod \Phi_r$. □

For $r = 5$, $d = 4$ this is Váňa's $\rho_5(p) \mid 10(p^2-1)$ [6], the fact that makes the L–P construction's cost the *double* condition ($p-1$ and $p+1$ smooth: dimension $1+1=2$) rather than the full $p^4 - 1$. The collapse exists because the zero locus $\{1\}$ of $X-1$ is stable under the Galois action, with root-of-unity cofactor. The second section has no such structure:

Lemma 3 (trivial stabilizer)**.** *For every $\gamma \in (\mathbb{Z}/r)^\times$ with $\gamma \neq 1$, the element $(\zeta^\gamma + 2)/(\zeta + 2)$ is not a unit of $\mathbb{Z}[\zeta_r]$ — in particular not a root of unity. Consequently no identity of the form $(X+2)^{p^e-1} \in \mu_{2r}$ holds for any e with $p^e \not\equiv 1 \pmod r$: the collapse of Lemma 2 has no analogue for $X+2$ at any depth.*

Proof. First, $N(\zeta+2) = \Phi_r(-2) = (2^r+1)/3$ (from $X^r - 1 = (X-1)\Phi_r(X)$ at $X = -2$), which is coprime to r: by Fermat $2^r \equiv 2 \pmod r$, so $2^r + 1 \equiv 3$, and $\gcd(r, 3) = 1$. Let $\mathfrak{q}$ be a prime of $\mathbb{Z}[\zeta_r]$ dividing $(\zeta+2)$; then $\zeta \equiv -2 \pmod{\mathfrak{q}}$, and the image of -2 in the residue field has multiplicative order exactly r (it is the image of ζ). If $\mathfrak{q}$ also divided $(\zeta^\gamma + 2)$ then $\zeta^\gamma \equiv -2 \equiv \zeta$, so $\zeta^{\gamma-1} \equiv 1 \pmod{\mathfrak{q}}$, forcing $r \mid \gamma - 1$. Hence for $\gamma \neq 1$ the ratio has a pole at $\mathfrak{q}$ and is not a unit. □

Remark 4 (two structural asides). (i) The $X = 1$ component of $T(2, n, r)$ is the base-3 Fermat condition $3^n \equiv 3 \pmod n$ — automatic for Carmichael-type n, but nontrivial, unlike the first section's $0 = 0$. (ii) Modulo $p = 3$ one has $-2 \equiv 1$, so the two sections *coincide*: the pair constrains only the divisors $p > 3$ of n. Finitely many p (those dividing $2^r + 1$) make $X+2$ a zero divisor and are likewise excluded.

3 The burden jump

Definition 5 (sharpness hypothesis)**.** *Say* generic sharpness *holds at (r, p) for a section f if every prime $\ell > 2rd$ dividing $\Phi_e(p)$ for a level e in the support permitted by the collapse structure actually divides* $\operatorname{ord}(f)$*. (Failure at ℓ requires the image of f to be an ℓ-th power in every component,*

heuristic probability $\leq 1/\ell$*; in our exact-order computations,* 0 *failures occurred in* 196 *sampled large primes across* $67 + 56$ *sample primes* p *and moduli* $r \leq 41$*.)*

Theorem 6 (Popovych burden)**.** *Let* n *be produced by the Frobenius-matching mechanism for the pair: for every* $p \mid n$ *there is* j_p *with* $n \equiv p^{j_p} \pmod{\operatorname{lcm}(r, \rho_r(p), \rho'_r(p))}$*, and* $\operatorname{ord}_r(n) > 2$ *(validity). Assume generic sharpness for both sections. Then for every* $p \mid n$*, writing* $d_p = \operatorname{ord}_r(p)$ *(and noting validity forces* $d_p \geq 3$*), the quantity* $\prod_{e \mid d_p} \Phi_e(p)$ *— a product of cyclotomic torus orders of total dimension* $\sum_{e \mid d_p} \varphi(e) = d_p$ *— must be smooth enough to divide the construction's common modulus. In particular the minimal per-prime burden over admissible cells is*

$$\operatorname{burden}_{\mathrm{Pop}}(r) \;=\; \min\{\, d : d \mid r-1,\; d \geq 3 \,\} \;\geq\; 3 \qquad \textit{for every prime } r,$$

whereas the single-section (Agrawal) mechanism has burden $\min_{d \mid r-1,\, d \geq 3}(d/2$ *if* d *even, else* $d)$*, with global optimum* 2*. At* $r = 5$*:* $\operatorname{burden}_{\mathrm{Pop}} = 4$*, i.e.* $p-1$*,* $p+1$ *and* p^2+1 *simultaneously smooth, against the single section's* 2*.*

Proof. For odd d_p both sections are supported on all levels $e \mid d_p$ (no involution in $\langle p \rangle$, and Lemma 3 forbids any other collapse); total dimension d_p. For even d_p the first section collapses to levels $e \mid d_p/2$ (Lemma 2) but by Lemma 3 and sharpness, $\rho'_r(p)$ retains the top level $e = d_p$, so $\operatorname{lcm}(\rho_r, \rho'_r)$ spans all $e \mid d_p$: the collapse is destroyed and the total is again d_p. Minimizing d_p over $d \mid r-1$, $d \geq 3$ gives the formula; the least such divisor is 3 iff $3 \mid r-1$, and for $r = 5$ (where $r - 1 = 4$) it is 4, with levels $\{1, 2, 4\}$ and $\Phi_4(p) = p^2 + 1$. The Agrawal comparison is Lemma 2 plus the same minimization. □

Corollary 7 (cost at representative parameters)**.** *For pool primes of* b *bits and smoothness bound* B*, the extra quadratic level of Theorem 6 at* $r = 5$ *costs a Dickman factor* $\rho(2b/\log_2 B)$ *beyond the double condition. At the parameters of the Agrawal-directed harvest study in this repository (*$b = 51$*;* $B = 1259$ *for the solver base,* $B' = 12590$ *for the enumerated base), this is* $\rho(9.9) \approx 3.9 \times 10^{-11}$*, respectively* $\rho(7.5) \approx 1.8 \times 10^{-7}$ *— reducing that study's measured pool of* 2.5×10^5 *elements to between* 10^{-5} *and* 0.05*: empty at any feasible scale. Unlike* $p \mp 1$*, where partition-first harvesting lets one side be enumerated rather than tested,* $\Phi_4(p)$ *is pinned once* p *is, so no enumeration trick removes the tax.*

r	Agrawal burden	Popovych burden	Δ
5	2	4	+2
7	3	3	0
11	5	5	0
13	2	3	+1
17	2	4	+2
19	3	3	0
23	11	11	0
29	2	4	+2
31	3	3	0
37	2	3	+1
41	2	4	+2

The deltas vanish exactly at $r \equiv 3 \pmod 4$, where the Agrawal burden is already ≥ 3 and both conjectures share full depth; the Agrawal global optimum (2, at any $r \equiv 1 \bmod 4$) is unavailable to the pair at every modulus.

Remark 8 (the constant 2 is immaterial). For any integer $c \notin \{0, \pm 1\}$, Lemma 3 and Theorem 6 hold verbatim for the pair $\{X-1, X+c\}$: the proof needs only that at a prime $\mathfrak{q} \mid (\zeta+c)$ the image of $-c$ has multiplicative order r, which holds at the primitive prime divisors of $N(\zeta+c) = \Phi_r(-c) = (c^r+1)/(c+1)$. The constant enters only through that norm (its primes, with $p \mid c(c+1)$, are the finitely many excluded p), the excluded modulus $r \mid c+1$ (for $c=2$: our standing $r \neq 3$), and the level-one component, the base-$(c+1)$ Fermat condition of Remark 4. What must be avoided is exactly degeneracy of the zero $-c$: it must be a point of $\mathbb{G}_m$ avoiding the rational torsion $\{\pm 1\}$, i.e. $c \notin \{0, \pm 1\}$. The exclusion $c=0$ is vacuous rather than subtle: X is a monomial, so $(X)^n \equiv X^n$ is an identity in every ring (the zero sits at the origin, off the torus, and the section is a unit of finite order r). Among nonzero c: $c=-1$ is the first section itself, and $c=+1$ — the one innocent-looking bad choice — has its zero at -1, fixed by inversion, so the collapse of Lemma 2 *survives* for $X+1$: $(\zeta+1)^{p^{d/2}} = \zeta^{-1}+1 = \zeta^{-1}(\zeta+1)$, whence $\mathrm{ord}(X+1) \mid r(p^{d/2}-1)$ for even d; moreover $\zeta+1 = (\zeta^2-1)/(\zeta-1)$ is multiplicatively dependent on the Galois orbit of $\zeta-1$, so $\{X-1, X+1\}$ adds no rank and no burden: a Popovych-style conjecture with $c=1$ would be no harder for the mechanism than Agrawal's. Thus $c=2$ is simply the smallest valid choice ($c=-2$ works equally, with Mersenne norm 2^r-1). Numerically (`code/ag5b_constant_check.py` of the repository): at even d the section $X+1$ collapses in 22/22 samples with the top level never needed, while $c \in \{2, 3, -2, 5\}$ collapse only at the single degenerate point $(r,p) = (13,5)$ of Section 4.

Proposition 9 (the $X+1$ variant is mechanism-equivalent to Agrawal's conjecture)**.** *Suppose* $n \equiv p^j \pmod{\mathrm{lcm}(r, \rho_r(p))}$ *and* $p-1 \mid n-1$ *(the mechanism congruence together with the Carmichael condition, both imposed by every L–P-style construction). Then* $T(1,n,r)$ *holds modulo* p *outright. Consequently every counterexample to Agrawal's conjecture produced by the mechanism also defeats the variant conjecture whose hypothesis is the pair* $\{T(-1,n,r), T(1,n,r)\}$*: that variant, though formally stronger-looking, is exactly as resistant as Agrawal's — the second hypothesis is free armor — whereas* $c=2$ *(or any* $c \notin \{0, \pm 1\}$*) genuinely blocks the engine.*

Proof. At the $X=1$ component, $(1+1)^n = 2^n \equiv 2 \pmod{p}$ since $\mathrm{ord}_p(2) \mid p-1 \mid n-1$. In the Φ_r components, $\zeta+1 = (\zeta^2-1)/(\zeta-1)$, a unit (its norm is $\Phi_r(-1)=1$), and both $\zeta-1$ and ζ^2-1 are values of $X-1$ at primitive r-th roots of unity, so their orders divide $\rho_r(p) = \mathrm{lcm}_a \,\mathrm{ord}(\zeta^a-1)$. Hence $(\zeta^a-1)^n = (\zeta^a-1)^{p^j} = \zeta^{ap^j}-1$ for $a=1,2$, and

$$(\zeta+1)^n \;=\; \frac{(\zeta^2-1)^n}{(\zeta-1)^n} \;=\; \frac{\zeta^{2p^j}-1}{\zeta^{p^j}-1} \;=\; \zeta^{p^j}+1 \;=\; \zeta^n+1,$$

using $n \equiv p^j \pmod{r}$ in the last step. Measured (`code/ag5b_constant_check.py`, check W): 48/48 synthetic mechanism instances pass $T(1,n,r)$, while 0/48 pass $T(2,n,r)$. □

4 Numerical verification

All claims were verified by a deterministic script (`code/ag5_popovych_check.py` of the repository, with committed JSON output), on 56 sample primes spanning $(r,d) \in \{(5,4),(13,4),(41,4),(7,6),(7,3),(11,5),(23$ Frobenius matching for $X+2$ (56/56); the first section's collapse bound fails to absorb $X+2$ at even d (35/36); the top level $e=d$ is present in $\rho'_r(p)$ (55/56); large primes of all levels $e \mid d$ divide $\rho'_r(p)$ (47/47, ≈ 0.28 misses expected under the $1/\ell$ model); $\gcd(r, 2^r+1) = 1$ at all tested r. The single exception in the second and third checks is $r=13$, $p=5$, where $\Phi_4(5) = 26 = 2 \cdot 13$ contains no prime outside $2r$ at all — a vacuous top level at one tiny prime, consistent with (indeed predicted by) the generic model, not a collapse event.

5 Scope

Three boundaries. (i) *Mechanism scope*: the results quantify Frobenius-matching constructions, the family containing every published counterexample approach [3, 4, 7]; "accidental" solutions of the component equations outside it are not excluded, and no way to construct them is known. (ii) *Conditionality*: the burden *upper* structure (the collapse identities and support bounds) is unconditional; the *lower* bounds use generic sharpness (Definition 5), an Artin-type hypothesis — a construction could in principle fish in the thin set of primes whose section orders collapse further, of heuristic density $\sim 1/\ell$ per avoided large prime, far thinner than the pools in question. (iii) *Direction*: this is evidence that the only known counterexample engine is strictly more blocked against Conjecture 1 than against Agrawal's conjecture — a reason the conjecture has resisted the L–P heuristic — not evidence of its truth in any stronger sense.

References

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